

AI Learning and Course Management Platform

A PROJECT REPORT

Submitted by

Shraddhaba Digvijaysinh Rana

230263107029

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER ENGINEERING



**KALOL INSTITUTE OF TECHNOLOGY & RESEARCH
CENTRE, KALOL**

At- Kalol

May 2026-27



CERTIFICATE

This Is To Certify That The Internship Report Submitted Along With The Project Entitled **AI Learning and Course Management Platform** Has Been Carried Out By **Shraddhaba Digvijaysinh Rana** Under My Guidance In Partial Fulfillment For The Degree Of Bachelor Of Engineering In Computer Engineering, 8th Semester Of Kalol Institute Of Technology & Research Centre, Kalol During The Academic Year 2026-27.

Prof. Mr. Saumil Barot
Internal Guide

Prof. . Mrs Tejal Purani
Head of Department



Faculty of Engineering

DECLARATION

I hereby declare that the Internship Report submitted along with the Internship Project entitled “**AI Learning and Course Management Platform**” in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering to **Kalol Institute of Technology & Research Centre, Kalol**, is a Bonafide record of original work carried out by me during the internship.

I further declare that this project work has been completed under the guidance of **Prof. Mr. Saumil Barot**, and has not been submitted elsewhere for the award of any degree or diploma. No part of this report has been directly copied from any other student's work or from any external source without proper acknowledgment and reference.

Shraddhaba Digvijaysinh Rana

ACKNOWLEDGMENT

I would like to express my sincere gratitude to all those who have contributed to the successful completion of this project titled “**AI Learning and Course Management Platform.**”

First and foremost, I am deeply thankful to my project guide, **Mr. Saumil Barot**, for their continuous support, valuable guidance, and encouragement throughout the development of this project. Their insights and suggestions played a crucial role in shaping the direction and quality of the work.

I would also like to extend my gratitude to the Head of the Department, **Mrs Tejal Purani**, and all the faculty members of the **Computer Engineering, Kalol Institute of Technology & Research Centre, Kalol**, for providing the necessary resources and academic environment to carry out this project successfully.

I am grateful to **VEER COMPUTING & CONSULTANCY, Ahmedabad.** for giving me the opportunity to work on this project and for their support during the internship/project duration.

I would also like to thank my teammates and peers for their cooperation, support, and constructive feedback throughout the project development process.

Finally, I express my heartfelt gratitude to my family for their constant encouragement and support, which motivated me to complete this project successfully

**Yours Sincerely,
Shraddhaba
Digvijaysinh Rana**

ABSTRACT

The rapid advancement of digital technologies has transformed traditional education systems into more flexible and accessible digital learning environments. Institutions and organizations increasingly require efficient systems to manage courses, students, and learning resources. This project presents the development of an AI Learning and Course Management Platform, a web-based system designed to integrate academic learning, online courses, and business training within a unified environment.

The proposed system addresses key challenges in existing manual and semi-digital systems, such as inefficient course management, difficulty in tracking student performance, lack of centralized resources, and absence of intelligent decision-making support. The platform provides features including course management, student enrollment, webinar management, resource sharing, and performance tracking.

A major contribution of this system is the integration of Artificial Intelligence to enhance learning outcomes. AI modules provide personalized course recommendations based on student behavior, performance analysis through data-driven insights, automated webinar summaries, and an intelligent learning assistant to guide students.

The system is developed using modern web technologies such as SvelteKit for frontend and backend integration, PostgreSQL/SQLite for database management, and AI APIs for intelligent features. The platform supports multiple user roles including Admin, Instructor, Student, and Business Manager, each with defined access control.

The proposed solution improves learning efficiency, enhances engagement, and provides scalable infrastructure for both educational institutions and corporate training environments. This project demonstrates how AI-driven systems can revolutionize learning management by combining automation, analytics, and user-centric design.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
LMS	Learning Management System
API	Application Programming Interface
UI	User Interface
UX	User Experience
DB	Database
ER	Entity Relationship
DFD	Data Flow Diagram
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UAT	User Acceptance Testing
SQL	Structured Query Language

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INTERNSHIP COMPLETION LETTER

Date: 20 April 2026
Ref No.: VCC/INT/2026/055

To,
The Principal / Head of Department,

Subject: Course Completion Certificate – Artificial Intelligence

Dear Sir/Madam,

This is to certify that Shraddhaba Digvijaysinh Rana (Enrollment No.: 230263107029) from Kalol Institute of Technology, Kalol has successfully completed the Artificial Intelligence Course at VEER COMPUTING & CONSULTANCY, Ahmedabad.

The course was conducted from 19 January 2026 to 20 April 2026, during which the student gained practical knowledge and hands-on experience in Artificial Intelligence concepts and applications.

During this period, the student gained practical knowledge in:

Fundamentals of Artificial Intelligence

Prompt Engineering & AI Tools

Chatbot Development (No-Code)

Real-world AI Applications

Productivity using AI

The performance of the student was found to be Excellent.

We wish them all the best for their future endeavors.

Warm regards,

Founder & IT Head



+91 92 65 65 2119

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Branch 1: Uttamnagar, Ahmedabad
Branch 2: Nikol, Ahmedabad

CHAPTER 1 – INTRODUCTION TO INTERNSHIP PROJECT

1.1 Project - Internship Summary

The project titled “**AI Learning and Course Management Platform**” is a web-based application designed to manage and enhance digital learning environments. The system integrates course management, student enrollment, webinar handling, resource sharing, and AI-driven features into a unified platform.

The primary goal of the system is to simplify learning management while improving user experience through intelligent automation and data-driven insights. The platform supports multiple user roles including Admin, Instructor, Student, and Business Manager.

1.2 Purpose

The purpose of this project is to develop a centralized and intelligent system that addresses the limitations of traditional and existing digital learning platforms. The system aims to improve efficiency in managing educational content and provide enhanced learning support using AI technologies.

1.3 Objectives

- To design a scalable web-based learning management system
- To enable efficient course and user management
- To provide online learning support through webinars
- To integrate AI features such as recommendations and performance analysis
- To improve user engagement and learning outcomes

1.4 Scope

The system is intended for educational institutions, training organizations, and businesses. It allows users to create, manage, and access learning content efficiently.

Scope Includes:

- Course creation and management
- Student enrollment and tracking
- Webinar scheduling and participation
- Resource management (documents, videos, etc.)
- AI-based recommendations and analytics

Limitations:

- Requires internet connectivity
- AI accuracy depends on available data
- Limited third-party integrations in current version

1.5 Technology

The system is developed using modern technologies to ensure scalability and performance:

- **Frontend & Backend:** SvelteKit
- **Database:** PostgreSQL / SQLite
- **Authentication:** Role-based access system
- **AI Integration:** External AI APIs

Existing systems such as traditional LMS platforms provide basic functionalities but lack advanced AI-driven features. This project builds upon those systems by incorporating intelligent automation and analytics.

1.6 Project Planning

1.6.1 Development Approach and Justification

The project follows an **Agile development approach**, allowing iterative development, continuous testing, and flexibility in incorporating new features. This approach is suitable due to the evolving nature of requirements and integration of AI components.

1.6.2 Effort, Time, and Cost Estimation

The project development is divided into phases including requirement analysis, design, development, testing, and deployment.

- Requirement Analysis: 1–2 weeks
- System Design: 2 weeks
- Development: 4–6 weeks
- Testing & Debugging: 2 weeks

Cost is minimized as the system uses open-source technologies and APIs.

1.6.3 Roles and Responsibilities

- **Developer:** System design, coding, and implementation
- **Designer:** UI/UX design and user experience optimization
- **Tester:** Testing functionality and identifying bugs
- **Project Lead:** Planning, coordination, and decision-making

1.6.4 Dependencies

- Internet connectivity
- Availability of AI APIs
- System compatibility with browsers and devices

1.7 Project Scheduling

The project follows a structured timeline where each phase is completed sequentially with some overlap in development and testing phases. Agile sprints are used to track progress and ensure timely completion of tasks.

CHAPTER 2 – SYSTEM ANALYSIS

2.1 Study of Current System

The current learning and course management systems used by many institutions are either manual or partially digital. Traditional systems rely on paperwork and offline coordination, resulting in inefficiencies, redundancy, and difficulty in accessing data.

Existing digital systems provide limited functionalities and often operate in isolation, lacking integration between modules such as course management, student tracking, and resource sharing. Moreover, these systems do not incorporate intelligent features to enhance learning outcomes.

2.2 Problems in Current System

The major limitations of existing systems include:

- Manual handling of course and student data
- Difficulty in tracking student progress and performance
- Lack of centralized resource management
- Limited support for webinars and live learning
- Absence of AI-based insights and automation
- Poor scalability and flexibility

2.3 Requirements of New System

Functional Requirements

- User registration and role-based authentication
- Course creation and management
- Student enrollment system
- Webinar scheduling and participation
- Resource upload and access
- Performance tracking
- AI-based recommendations
- AI learning assistant

Non-Functional Requirements

- Scalability
- Security
- Reliability
- User-friendly interface
- Fast response time

2.4 System Feasibility

2.4.1 Organizational Objective

The system can be implemented using SvelteKit for full-stack development, PostgreSQL/SQLite for database management, and AI APIs for intelligent features.

2.4.2 Technical & Cost Feasibility

The use of open-source technologies minimizes development and operational costs.

2.4.3 Integration

The system is user-friendly and can be adopted easily by institutions and organizations.

2.5 Process in Proposed System

The proposed system is an **AI Learning and Course Management Platform** that integrates course management, student tracking, resource handling, and AI-driven functionalities into a unified system.

2.6 Use Case Diagram

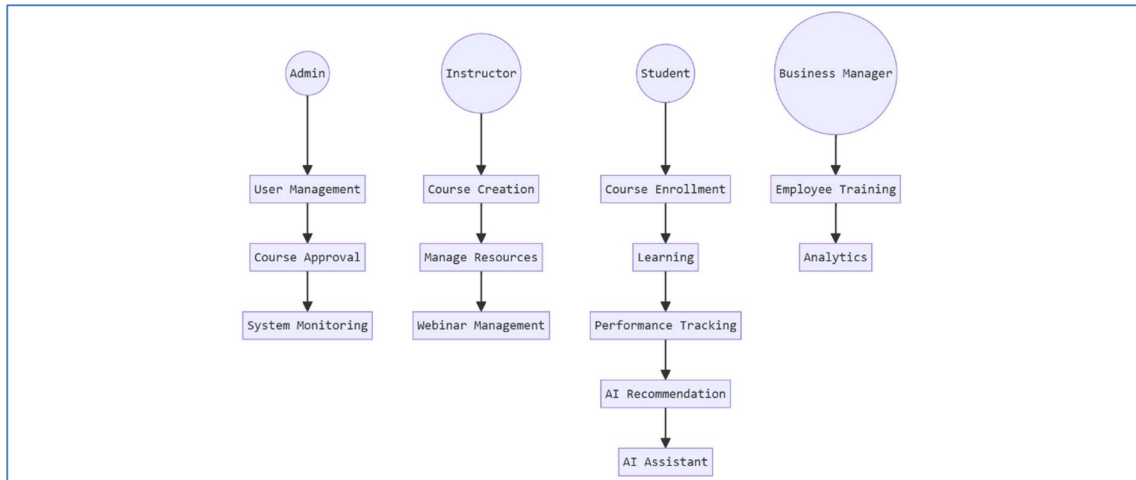


Fig 2.1 Use Case Diagram

This diagram represents interactions between different user roles and the system functionalities, showing how each user engages with the platform.

2.7 Activity / Process of Proposed System

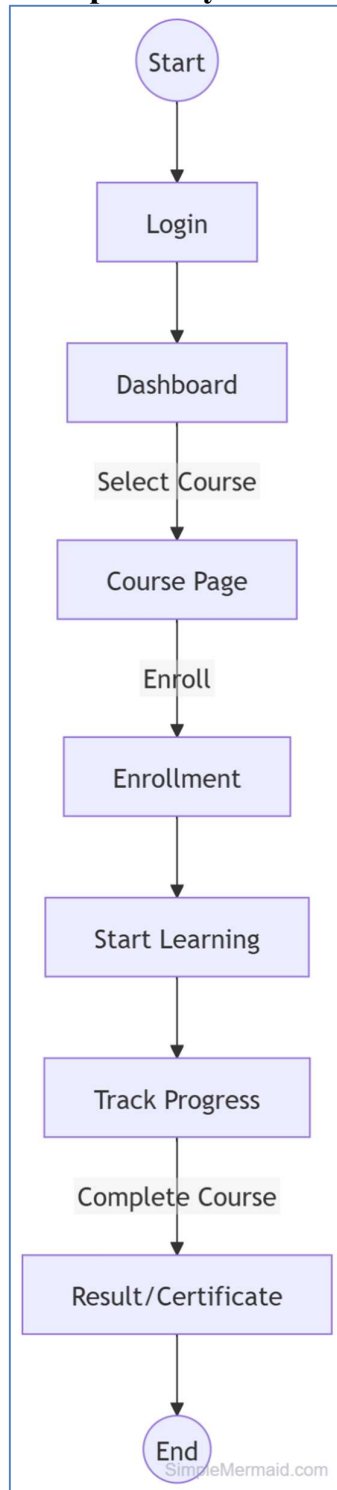


Fig 2.2 Activity Diagram

This diagram illustrates the workflow of a student from login to course completion.

2.8 Data Flow Diagram (DFD - Level 1)

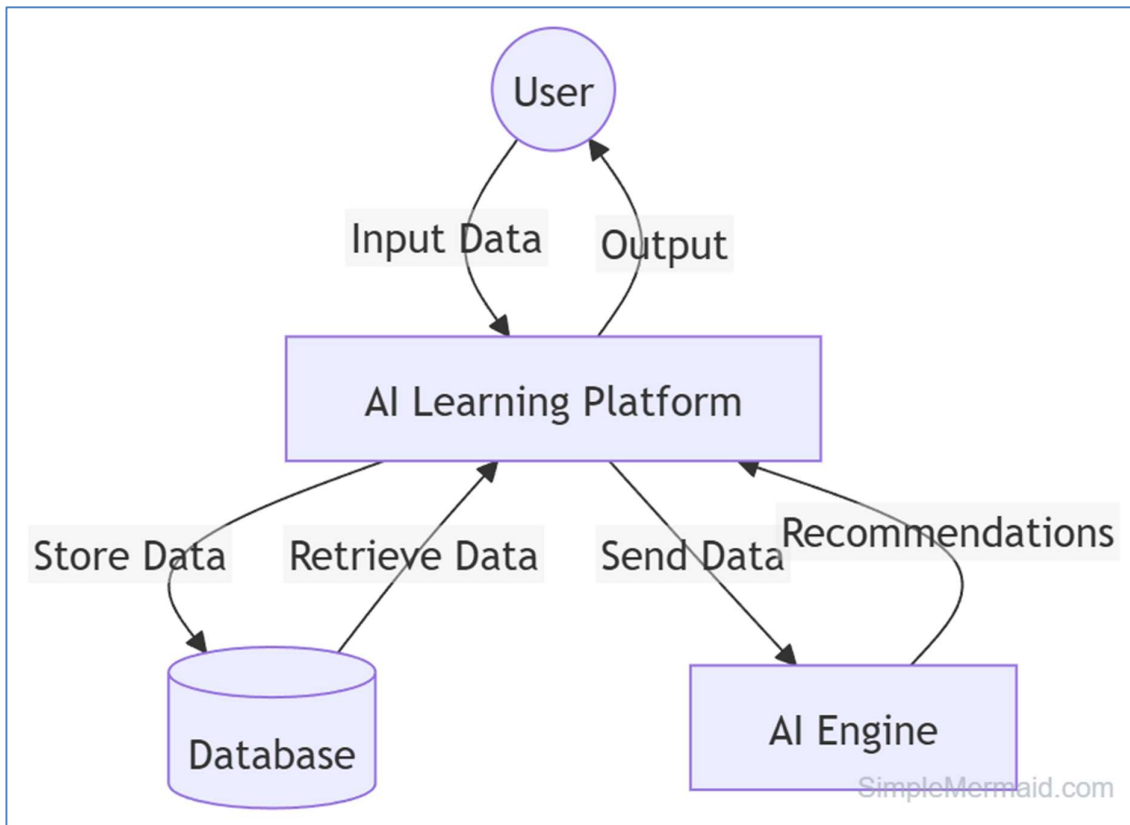


Fig 2.3 Data Flow Diagram

This diagram shows how data flows between users, system components, database, and AI modules.

2.9 Features of Proposed System

- Centralized course and user management
- Seamless student enrollment
- Webinar scheduling and management
- Resource library
- Performance tracking and analytics
- AI-based recommendations
- AI learning assistant

2.10 System Modules

The system is divided into the following modules:

- Admin Module
- Instructor Module
- Student Module
- Business Training Module
- AI Module

2.11 Selection of Technology and Justification

The technologies are selected based on scalability, performance, and flexibility:

- SvelteKit for full-stack development
- PostgreSQL/SQLite for database management
- AI APIs for intelligent features
- Web-based architecture for cross-platform accessibility

These technologies ensure that the system is efficient, scalable, and future-ready.

CHAPTER 3 – SYSTEM DESIGN

3.1 System Design & Methodology

The system is designed using a modular and layered architecture to ensure scalability, maintainability, and performance. A client-server model is followed, where users interact through a web interface, and processing is handled on the server.

The development follows an Agile methodology, enabling iterative improvements, continuous testing, and flexibility in integrating AI features.

3.2 System Architecture Design

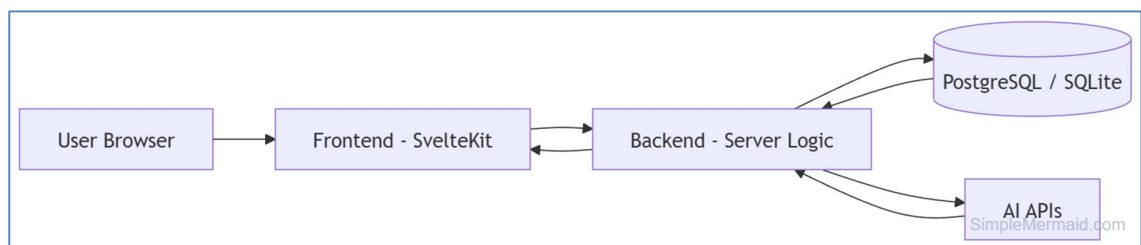


Fig 3.1 System Architecture Diagram

This diagram represents the three-tier architecture of the system, where the frontend handles user interaction, the backend processes logic, and the database stores data. AI APIs are integrated to provide intelligent functionalities.

3.3 Database Design (ER Diagram)

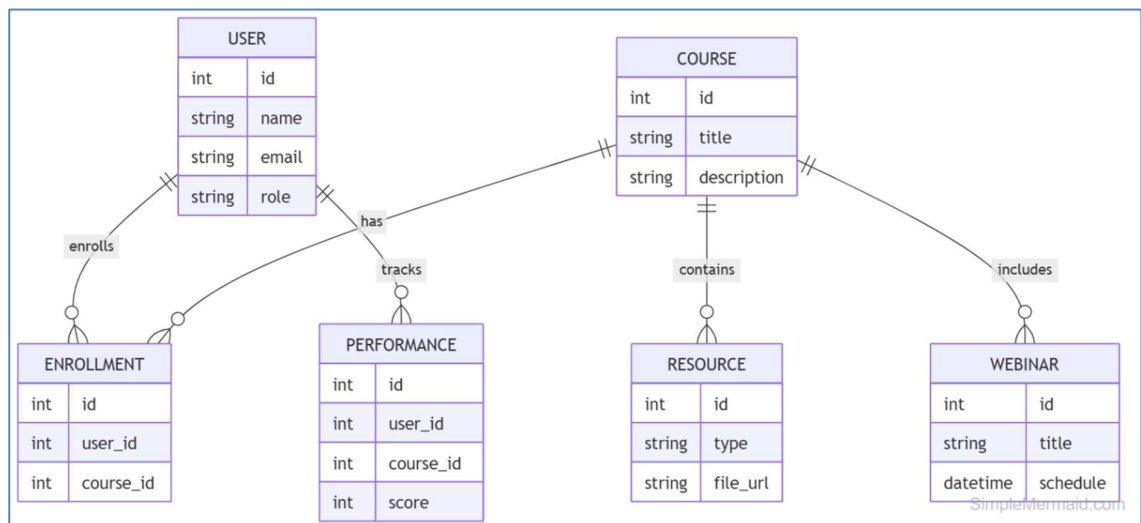


Fig 3.2 ER Diagram

This diagram defines the database structure and relationships between entities such as users, courses, enrollments, resources, webinars, and performance records.

3.4 Data Structure Design

The system uses structured relational data models:

- Primary keys ensure unique identification of records
- Foreign keys maintain relationships between entities
- Indexed fields improve query performance
- Normalization reduces redundancy and ensures data consistency

3.5 Input and Output Design

Input Design

Inputs to the system include:

- User registration and login data
- Course creation details
- Enrollment requests
- Resource uploads
- Webinar scheduling

All inputs are validated to prevent incorrect or malicious data entry.

Output Design

Outputs generated by the system include:

- Course listings and details
- Student progress reports
- Webinar schedules and summaries
- AI-generated recommendations
- Performance analytics dashboards

3.6 Interface Design

The system provides role-based dashboards:

- Admin Dashboard: User and system management
- Instructor Dashboard: Course creation and content management
- Student Dashboard: Learning and progress tracking
- Business Dashboard: Employee training monitoring

The interface is designed to be responsive, user-friendly, and accessible across devices.

3.7 Security and Access Control

Security is implemented using role-based access control mechanisms:

- Authentication using secure login credentials
- Authorization based on user roles
- Data validation and sanitization
- Protection against unauthorized access

3.8 Design Constraints

The system design is subject to the following constraints:

- Requires stable internet connectivity
- Performance depends on server capacity
- AI features depend on external APIs
- Cross-browser compatibility must be maintained

3.9 Design Justification

The chosen architecture and technologies ensure:

- Scalability for handling multiple users
- Efficient data management
- Flexibility for future enhancements
- Integration of AI without heavy infrastructure

This design provides a balance between performance, cost, and maintainability.

3.10 Project UI

This section includes actual output screenshots of:

- Login Page
- Dashboard
- Course Page
- AI Recommendation Output
- Performance Dashboard



AI Learning Platform

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Student ID: #9921

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65% Completed • 45 mins left in this module

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AI-Powered Recommendations

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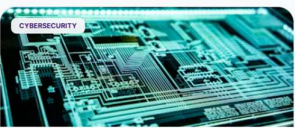


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Dashboard Overview

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Total Students

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+12%

Monthly Revenue

\$14,250

+8%

Completion Rate

84.2%

Stable

Avg. Instructor Rating

4.88

4.9/5

Active Courses

COURSE NAME	STATUS	STUDENTS	REVENUE	ACTION
Advanced Prompt Engineering Last updated 2 days ago	Published	1,240	\$8,420	Edit Course
AI in FinTech: Masterclass Created March 15, 2026	Published	656	\$5,210	Edit Course
Cybersecurity for AI Models Draft saved 1 hour ago	Draft	0	\$0	Continue Editing

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PREVIEW

Module 1: Introduction to Transformer Architecture

14:20

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Master the core concepts of Generative AI. This comprehensive course covers everything from basic LLM fine-tuning to deploying production-ready diffusion models for image generation.

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🌐 English, Spanish

Your Instructor

Dr. Sarah Chen

Head of AI Research @ Nexus Labs

Sarah is a pioneer in NLP research with over 15 years of experience. She has led AI teams at Google and DeepMind, focusing on scaling transformer-based models.

Reviews

Write a review

Marcus Thorne

★★★★★

The explanation of Attention mechanisms was the clearest I've ever seen. Highly recommended for engineers moving into AI.

Course Content

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MODULE 1: FOUNDATIONS

Introduction to Transformers

14:20

Attention is All You Need

22:05

Encoders vs Decoders

18:12

MODULE 2: LARGE LANGUAGE MODELS

Pre-training Techniques

45:00

Fine-tuning with RLHF

1h 12m

MODULE 3: VISUAL GENAI

Stable Diffusion Internals

55:20

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22

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Your Intelligence Profile

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88%

CREATIVE PROMPTING

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REAL-TIME SKILL MAPPING

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Certificates

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92.4% + 4.1%

STUDY HOURS (MONTHLY)

42h 15m + 12%

MODULES COMPLETED

18 / 24 ON TRACK

Learning Activity

Last 7 Days

Skill Distribution

TOTAL 88%

Technical 65%

Conceptual 23%

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Deep Learning Specialization 85% Complete

Applied Generative AI 42% Complete

Natural Language Processing Completed

23

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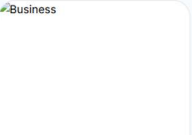
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Certificate

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AI Assistant

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AI Learning Tutor

GPT-4 TURBO INTEGRATED

Hello Alex! I see you were recently studying **Module 4: Neural Networks**. How can I assist you today? I can explain complex concepts, help with code snippets, or quiz you on the recent material.

AI ASSISTANT • 10:30 AM

Can you explain "Backpropagation" simply? Use a real-world analogy if possible.

YOU • 10:32 AM

Think of **Backpropagation** like a chef trying to perfect a soup recipe:

- The Taste (Forward Pass):** You make the soup and taste it. It's too salty.
- The Error (Loss Calculation):** The "error" is the difference between your current salty soup and the perfect soup.
- The Correction (Backprop):** You go backward through your steps—too much salt at the end? Or too much bouillon at the start? You adjust the ingredients for next time.

```
# Conceptual Gradient Update
loss = criterion(output, target)
loss.backward() # The magic happens here!
optimizer.step()
```

AI ASSISTANT • 10:33 AM

Summarize Lesson

Explain Code

Quiz Me

Ask your AI tutor anything...

AI CAN MAKE MISTAKES. CHECK IMPORTANT INFO.

Alex Rivers

Free Tier

Resource Library

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2.4 MB

GPT-4 Training Insights

Expert interview recording from the 2026 AI Summit.

Play Video

45:12 MIN

Financial Forecast Dataset

Cleaned CSV and JSON files for the Capstone Project.

PREMIUM

1.2 GB

Python Prompt Templates

Cheat sheet for structuring LLM prompts efficiently.

View Details

850 KB

Showing 4 of 124 files

<

1

2

>

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MAY 02 SAT

Prompt Engineering for UI/UX Designers

Sarah Jenkins

4:00 PM EST

[Remind Me](#)

MAY 05 TUE

Career Pathing in the Age of Artificial General Intelligence

David Chen

10:00 AM EST

[Remind Me](#)

Past Recordings

Ethics in AI Data Labeling

Recorded April 20, 2023

Cybersecurity for AI Models

Recorded April 18, 2023

Large Language Models Training Loop

Recorded April 15, 2023

Building with Stable Diffusion API

Recorded April 12, 2023

[View All](#)

CHAPTER 4 – IMPLEMENTATION

4.1 Implementation Environment

The system is developed using modern web technologies to ensure performance, scalability, and maintainability.

- Frontend & Backend: SvelteKit
- Database: PostgreSQL / SQLite
- Programming Languages: JavaScript, HTML, CSS
- AI Integration: External AI APIs
- Development Tools: VS Code, Git
- Browser Support: Chrome, Edge, Firefox

The implementation environment supports full-stack development and efficient deployment.

4.2 System Modules Implementation

The system is implemented in a modular manner, where each module performs specific functionalities.

4.2.1 Admin Module

- Manage users and roles
- Approve or remove courses
- Monitor system activities

4.2.2 Instructor Module

- Create and manage courses
- Upload learning materials
- Schedule webinars

4.2.3 Student Module

- Register and log in
- Enroll in courses
- Access learning materials
- Track progress

4.2.4 Business Training Module

- Manage employee training programs
- Track employee performance

4.2.5 AI Module

- Course recommendation system
- AI learning assistant
- Performance analysis

4.3 Process and Technology Implementation

The system follows a client-server model:

- Users interact through the frontend interface
- Requests are processed by the backend server
- Data is stored and retrieved from the database
- AI APIs process data to generate intelligent outputs

The implementation ensures smooth communication between all components.

4.4 Key Functionalities Implemented

- User authentication and role management
- Course creation and enrollment system
- Resource management (documents, videos)
- Webinar scheduling and participation
- Performance tracking dashboard
- AI-based recommendations and assistant

4.5 Results and Outputs

The system successfully performs all major operations:

- Efficient management of courses and users
- Seamless enrollment process
- Real-time access to learning resources
- Improved tracking of student performance
- AI-generated insights and recommendations

4.6 Result Analysis

The implemented system improves efficiency compared to traditional systems:

- Reduces manual effort
- Enhances user experience
- Provides intelligent insights
- Supports scalability

The integration of AI significantly improves personalization and decision-making.

4.7 Limitations in Implementation

- AI features depend on external APIs
- System performance depends on server capacity
- Limited offline functionality
- Advanced analytics features can be improved

4.8 Future Improvements in Implementation

- Integration of mobile application
- Advanced AI models for better recommendations
- Improved analytics dashboard
- Integration with external platforms

CHAPTER 5 – TESTING

5.1 Testing Strategy

The system is tested using a combination of manual and functional testing approaches to ensure that all modules operate correctly and meet the specified requirements.

The testing process includes:

- Unit Testing for individual components
- Integration Testing for module interaction
- System Testing for overall functionality
- User Acceptance Testing (UAT) for real-world usability

5.2 Testing Plan

The testing plan focuses on validating all major functionalities of the system, including user authentication, course management, enrollment, resource access, and AI features.

Each module is tested under different conditions to ensure reliability, accuracy, and performance.

5.3 Test Cases

Test ID	Test Condition	Expected Output	Actual Output	Result
TC01	User login with valid credentials	Successful login	Successful login	Pass
TC02	User login with invalid credentials	Error message	Error displayed	Pass
TC03	Course creation by instructor	Course created	Course created	Pass
TC04	Student enrollment	Enrollment successful	Enrollment successful	Pass
TC05	Resource upload	File uploaded	File uploaded	Pass
TC06	Webinar scheduling	Webinar scheduled	Webinar scheduled	Pass
TC07	AI recommendation	Relevant courses suggested	Courses suggested	Pass
TC08	Performance tracking	Progress displayed	Progress displayed	Pass

Table 5.3: Test Cases and Results

5.4 Test Results and Analysis

The test results indicate that all core functionalities of the system are working as expected. The system successfully handles user interactions, data processing, and AI-based operations without significant errors.

Minor issues encountered during testing were resolved during the debugging phase. The system demonstrates stability, reliability, and consistent performance.

5.5 Performance Testing

The system performs efficiently under normal usage conditions:

- Fast response time for user requests
- Smooth navigation between modules
- Efficient database operations
- Stable AI response generation

5.6 Security Testing

Security testing ensures that the system protects user data and prevents unauthorized access:

- Authentication validation
- Role-based access control verification
- Input validation to prevent invalid data
- Protection against unauthorized actions

5.7 Limitations of Testing

- Testing performed on limited datasets
- AI outputs may vary based on input data
- Load testing for large-scale users is limited

5.8 Summary of Testing

The testing phase confirms that the system meets its functional and non-functional requirements. The platform is stable, secure, and ready for deployment with scope for future improvements.

CHAPTER 6 – CONCLUSION AND DISCUSSION

6.1 Overall Analysis of Project Viability

The AI Learning and Course Management Platform proves to be a viable and effective solution for modern digital learning environments. The system successfully integrates course management, student tracking, resource handling, and AI-driven features into a unified platform.

From a technical perspective, the use of modern web technologies ensures scalability, performance, and maintainability. Economically, the system is cost-effective due to the use of open-source tools. Operationally, the platform is user-friendly and can be adopted by educational institutions and organizations with minimal training.

6.2 Problem Encountered and Solutions

During the development of the system, several challenges were encountered:

- Integration of AI APIs:

Difficulty in handling responses and ensuring relevant outputs

Solution: Implemented structured data handling and filtering mechanisms

- Data Management Complexity:

Managing relationships between multiple entities

Solution: Designed normalized database schema with proper relationships

- User Role Management:

Ensuring secure and accurate access control

Solution: Implemented role-based authentication and authorization

- Performance Optimization:

Handling multiple requests efficiently

Solution: Optimized database queries and modularized backend logic

6.3 Summary of Project Work

The project involved the complete development lifecycle including requirement analysis, system design, implementation, and testing. The platform supports multiple user roles and provides functionalities such as course creation, enrollment, webinar management, resource sharing, and performance tracking.

The integration of AI enhances the system by providing personalized recommendations, intelligent assistance, and performance insights.

6.4 Limitations

Despite its advantages, the system has certain limitations:

- Dependence on internet connectivity
- AI accuracy depends on available data
- Limited advanced analytics in current version
- Limited integration with external platforms

6.5 Future Enhancements

The system can be further improved by:

- Developing a mobile application
- Implementing advanced AI-based personalized learning paths
- Enhancing analytics and reporting features
- Integrating with third-party platforms (e.g., payment systems, external LMS)
- Improving scalability for large-scale deployment

6.6 Final Conclusion

The developed system demonstrates how modern technologies and artificial intelligence can be combined to create an efficient and intelligent learning management platform. It improves both administrative processes and learning experiences, making it a valuable solution for educational institutions and organizations.

The project highlights the importance of integrating automation, data analysis, and user-centric design in building future-ready systems.

APPENDIX

Appendix A: Sample Screens (UI Representation)

The system includes multiple user interfaces for different roles:

- Admin Dashboard (User & System Management)
- Instructor Dashboard (Course Creation & Content Upload)s
- Student Dashboard (Learning Interface & Progress Tracking)
- Business Dashboard (Employee Training Management)

Appendix B: Sample Database Tables

User Table Example

ID	Name	Email	Role
1	John Doe	john@email.com	Student
2	Jane Smith	jane@email.com	Instructor

Table B.1

Course Table Example

ID	Title	Instructor
101	Web Development	Jane Smith
102	AI Basics	John Doe

Table B.2

Appendix C: Sample Code Snippet

```
// Example: User Authentication Logic
async function loginUser(email, password) {
  const user = await db.findUserByEmail(email);
  if (!user) return "User not found";

  if (user.password === password) {    return "Login Successful";
  } else { return "Invalid Credentials"; }
}
```

Appendix D: Test Case Details

Test ID	Input	Expected Output	Actual Output
TC01	Valid Login	Success	Success
TC02	Invalid Login	Error	Error

Table D.1

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